

STAT 391: FINAL PROJECT

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1. INTRODUCTION AND DATA

1.1. Goals. This research investigates music trends across time and genres by using various musical features, such as beats per minute (BPM), energy, and length, in songs that reached number one on the Billboard Hot 100 between August 4, 1958 and January 11, 2025. Our goal is to understand more about what makes a song successful, how these change over time, and how these factors are related to genres. To do this, we will answer three research questions:

1. What keeps a song at number one?²
2. Have song trends changed over time?³
3. What defines a genre?⁴

1.2. Dataset Description. To answer our research questions, we will be using the dataset Billboard Hot 100 Number Ones from <https://github.com/rfordatascience/tidytuesday/tree/main/data/2025/2025-08-26>[2]. This data consists of 1173 songs with 112 features each. However, we chose only 12 features to keep in our use of the data. The 12 we chose were song name, artist, date the song reached number one, genre (as assigned by dataset creator Chris Dalla Riva and Vinnie Christopher), the number of weeks the song spent at number one (including non-consecutive weeks), if the artist was a talent contestant, the keys used in the song, the simplified key used in the song, beats per minute (BPM), energy (scale from 0 to 100), danceability (scale from 0 to 100), happiness (scale from 0 to 100), and length in seconds. We chose to keep song and artist to recognize songs if needed, such as determining outliers if any occurred. We kept date to compare trends across time and similarly genre to compare trends across genres. The rest were kept as the possible factors that answer our research questions that we were most interested in. Although the data contained many other predictors, it was not feasible for us to use all of them and many of them were not as well suited for our research questions as the ones we chose.

2. WHAT KEEPS A SONG AT NUMBER ONE?

2.1. Question. Every song in this dataset was a number one hit for at least one week, but we wondered if there were any factors that were correlated with the number of weeks that a song stayed number one. Investigating this question helps us to understand what features, if any, define a number one hit and can hint to if there are features that are more likely to make a song popular and keep it in the limelight for longer.

2.2. Methods. We used the number of weeks a song spent at number one as the response variable and made a full linear model which included most of the predictors in our data set: date, bpm, if the artist was a talent contestant, energy, danceability, happiness, length in seconds, simplified key, and the genre of the song. We also created an empty linear model consisting only of the intercept before using stepwise AIC-based selection to determine which factors to include in our final model. We tested the stepwise function allowing forward steps only and allowing steps in both directions.

Date: March 18, 2026.

After testing this, we realized that there were many overlaps in genres. For example, pop, pop; funk/soul, and funk/soul;pop were all recorded as different categories, and we thought that this may impact how significant genre appears in the model by overstating the number of categories, therefore making each category smaller. To combat this, We simplified the genre categories to only include pop, rock, hip hop, folk/country, funk/soul, and electronic/dance with each song only having one genre label instead of multiple.

We also recognized that the genre category had many NA values which required us to remove a portion of our data. We decided to test the same model selection method as above, but without the genre category so that we could add back in the rows that had to be omitted.

2.3. Results. In our initial test, both methods, forward and bidirectional stepping, concluded that the final model should only include the date and the energy of the song as no other factors improved the model's accuracy or were statistically significant. However, even though energy is included in the model, it has a p-value of 0.155 which is higher than our α of 0.05 so we do not consider it statistically significant. Additionally, the R-squared value of our model is extremely low, only 0.02287, leading us to conclude that our model does not explain much of the variability in our data and these factors cannot accurately determine how long a song will stay at number one.

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.714e-01  6.993e-02  11.031 < 2e-16 ***
## date        1.951e-10  3.728e-11   5.234 1.97e-07 ***
## energy      -1.612e-03  1.132e-03  -1.424   0.155
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7254 on 1172 degrees of freedom
## Multiple R-squared:  0.02287,    Adjusted R-squared:  0.0212
## F-statistic: 13.71 on 2 and 1172 DF,  p-value: 1.298e-06
```

FIGURE 1. Regression summary output from model created using AIC based model selection

Changing the genre labels did not change the results of the model as genre was still not a significant predictor for weeks at number one, date and energy were the only two factors included in the model, and energy's p-value was still too high to be considered statistically significant. The R-squared value did increase slightly, but at 0.04975, was still extremely small.

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.719e-01  6.910e-02  11.171 < 2e-16 ***
## date        3.230e-10  4.287e-11   7.533 1.04e-13 ***
## energy      -2.172e-03  1.129e-03  -1.923   0.0547 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7037 on 1084 degrees of freedom
## Multiple R-squared:  0.04975,    Adjusted R-squared:  0.04799
## F-statistic: 28.37 on 2 and 1084 DF,  p-value: 9.751e-13
```

FIGURE 2. Regression summary output from model with simplified genre classifications

Without including genre, we found that the date and energy were both still the only significant factors, and energy was not a significant factor.

Due to the extremely low R-squared values and limited number of predictors that improved the model's accuracies, we can conclude that there is not a consistent feature or set of features in songs that stay the number one song on the top 100 for longer periods of time. Instead, these differences in lasting popularity are likely due to song release patterns, musical trends at the time, or other features that would require testing beyond the dataset we are using for this project.

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.03621 -0.71839 -0.03652  0.50743  1.96957
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.714e-01  6.993e-02  11.031  < 2e-16 ***
## date        1.951e-10  3.728e-11   5.234  1.97e-07 ***
## energy      -1.612e-03  1.132e-03  -1.424    0.155
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7254 on 1172 degrees of freedom
## Multiple R-squared:  0.02287,    Adjusted R-squared:  0.0212
## F-statistic: 13.71 on 2 and 1172 DF,  p-value: 1.298e-06
```

FIGURE 3. Regression summary output from model without genre included.

3. HAVE SONG TRENDS CHANGED OVER TIME?

3.1. Question. Next, we wanted to test if the type of music that reaches number one has changed over time by looking at various predictors in the data, such as length in seconds, energy, danceability, and bpm.

3.2. Methods. To test this, we ran linear regression with various factors as the response variable and time as the predictor to see if the relationship would be considered statistically significant. When we tested with genre, we once again used our simplified genre labels from research question one 2.2.

3.3. Results. We found that length did change over time, but non-linearly. Instead, songs got longer in the middle of our dataset and were shorter on the two extremes, corresponding to the late-50s and the early 2020s 4a.

We also found date to be statistically significant predictor of danceability with songs becoming more "danceable" over time. However, the variability in danceability is high at all time periods so the relationship is not very strong and the R-squared value is low 4b.

Similarly, energy had a statistically significant relationship with time, but the R-squared value was extremely low and much lower than danceability, leading us to believe that time is not the main influence in how energetic a song is 4c.

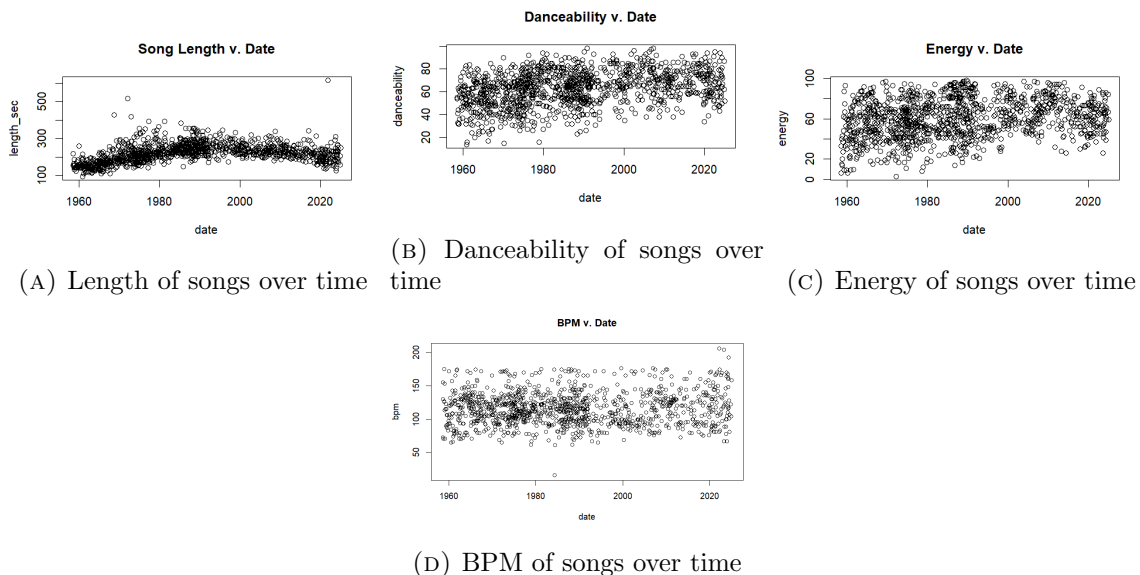


FIGURE 4. Plots for four of the tested features against time in years.

We found a statistically significant relationship between genre and date. Although the R-squared value is still low because many genres are represented in all periods of time, this allows us to conclude that the genres that are most popular have changed over time. To visualize this conclusion, we used ggplot to create a plot showing the number of songs of each genre every year[1]. As we can

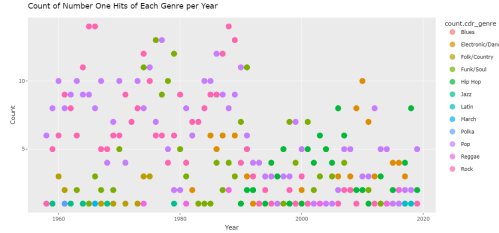


FIGURE 5. Plot showing number of songs of each genre per year.

see in this plot, although rock songs are represented throughout the entire time range, they were far more popular in the earlier years than in later years. In the opposite direction, electronic/dance music and hip hop are far more popular in recent years. Funk/soul also had a peak in the middle with few number one hits at either time extreme.

BPM remained relatively constant over time and although the coefficient is statistically significant relationship, the R-squared is extremely low, showing that most of the variability is not explained by time. This also seems clear looking at the plot which is mostly flat with no clear relationship 4d.

We can compare all p-values for the relationships between year and the features tested in a table.

feature	p-value
song length	$< 2e - 16$
danceability	$< 2e - 16$
energy	$< 2e - 16$
BPM	0.0412
genre	$< 2e - 16$

TABLE 1. p-values for the relationships between time and the features tested

4. WHAT DEFINES A GENRE?

4.1. Question. In our third research question, we explored what factors could be considered "genre defining" by seeing what factors were statistically significant predictors for the genre of a song. We considered the same factors as in the previous two questions: danceability, energy, song length, happiness, key, and bpm.

4.2. Methods. This question required us to use logistic regression as our output variable, genre, is categorical; however, we could still use methods similar to using linear regression. As with Question One2, we defined an empty and full logistic regression model then used a stepwise function with AIC to determine which variables were important in the model. This time, our full model used the factors listed above 4.1 as predictors. We once again tested once with forward stepping and once with two-directional stepping.

4.3. Results. The results of our model selection showed that danceability, happiness, energy, and bpm were all significant using a threshold of $\alpha = 0.05$. Some keys were also significant which shows that certain keys, such as Ab minor and C minor, are used more only in specific genres while others, like E and G, that were not significant are common across genres.

```

---
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    13.024264    0.878576  14.824 < 2e-16 ***
## danceability   -0.064798    0.007930  -8.171 8.67e-16 ***
## simplified_keyAb -1.239258    0.644905  -1.922 0.054923 .
## simplified_keyAbm -3.598398    0.923591  -3.896 0.000104 ***
## simplified_keyAm  -2.200249    0.676503  -3.252 0.001181 **
## simplified_keyB   -0.943225    0.763625  -1.235 0.217033
## simplified_keyBb  -0.887257    0.639281  -1.388 0.165460
## simplified_keyBbm -1.546836    0.849646  -1.821 0.068956 .
## simplified_keyBm  -2.169219    0.817826  -2.652 0.008111 **
## simplified_keyC   -0.474698    0.517365  -0.918 0.359074
## simplified_keyCm  -3.197924    0.748556  -4.272 2.11e-05 ***
## simplified_keyD   -0.636042    0.574291  -1.108 0.268319
## simplified_keyDb  -1.672816    0.608557  -2.749 0.006083 **
## simplified_keyDbm -2.639488    0.806580  -3.248 0.001200 **
## simplified_keyDm  -1.625087    0.774787  -2.097 0.036190 *
## simplified_keyE   -0.365253    0.584368  -0.625 0.532081
## simplified_keyEb  -1.506323    0.662082  -2.275 0.023098 *
## simplified_keyEbm -3.247472    0.778665  -4.171 3.29e-05 ***
## simplified_keyEm  -2.443741    0.791803  -3.086 0.002079 **
## simplified_keyF   -1.198816    0.564401  -2.124 0.033898 *
## simplified_keyFm  -3.809457    0.741733  -5.136 3.34e-07 ***
## simplified_keyG   -0.328166    0.551760  -0.595 0.552129
## simplified_keyGb  -1.557828    0.746118  -2.088 0.037045 *
## simplified_keyGbm -1.866804    0.865919  -2.156 0.031319 *
## simplified_keyGm  -2.249023    0.749789  -3.000 0.002767 **
## simplified_keyMultiple Keys -0.548400    0.536039  -1.023 0.306514
## happiness       -0.015670    0.005260  -2.979 0.002957 **
## energy          -0.013396    0.006042  -2.217 0.026821 *
## bpm             -0.008889    0.004300  -2.067 0.038932 *
## length_sec      -0.004008    0.002212  -1.812 0.070225 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

FIGURE 6. Logistic regression summary for model to classify genre based on various factors.

5. CONCLUSION

In conclusion, we used a dataset that contained the number one songs on the Billboard Hot 100 chart from August 1958 until January 2025 to understand more about music trends over time and across genres. A lack of statistically significant relationships in question one showed that how long a song remains at number one likely depends on factors outside of the song itself and would require greater inquiry into culture and societal shifts to understand 2.3. Experimenting with the relationship between various song factors and dates showed that song lengths rose before falling again, the danceability, energy, and bpm of songs all have a positive relationship with time, and although many genres have continued to have number one hits throughout all of the recorded years, many also had prime years 3.3. Finally, we determined that danceability, happiness, energy, and bpm all contributed to which genre a song would be classified as 4.3. This connection may also

contribute to the relationship between some of these factors and time as some genres, and therefore the features used in those genres, had clear peaks.

Although we did not see significant results for all of our questions, we did learn more about musical genres and trends over time.

REFERENCES

- [1] Plotly graphing library for ggplot2, 2026. <https://plotly.com/ggplot2/>.
- [2] C. D. Riva. Billboard hot 100 number ones, 2025. Data retrieved from <https://github.com/rfordatascience/tidytuesday/tree/main/data/2025/2025-08-26>.